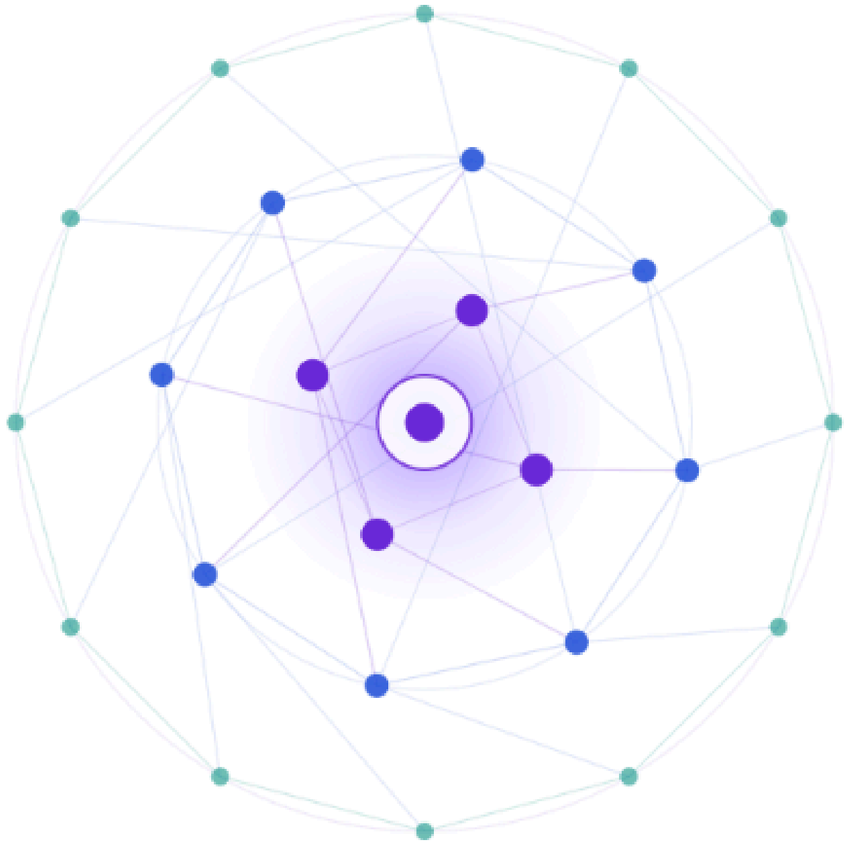
SAMSUNG PRISM

# Generative AI Hackathon

## 3rd Edition 2026 – 27



**Theme ID -** 02 · Smart Guided Troubleshooting Engine

**Team Name -** TIET_NEXORA

**College Name -** Thapar Institute of Engineering and Technology

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**Submission Github link -** https://github.com/smritixx/samsung-troubleshooting-engine

# Smart Guided Troubleshooting Engine

**Problem**

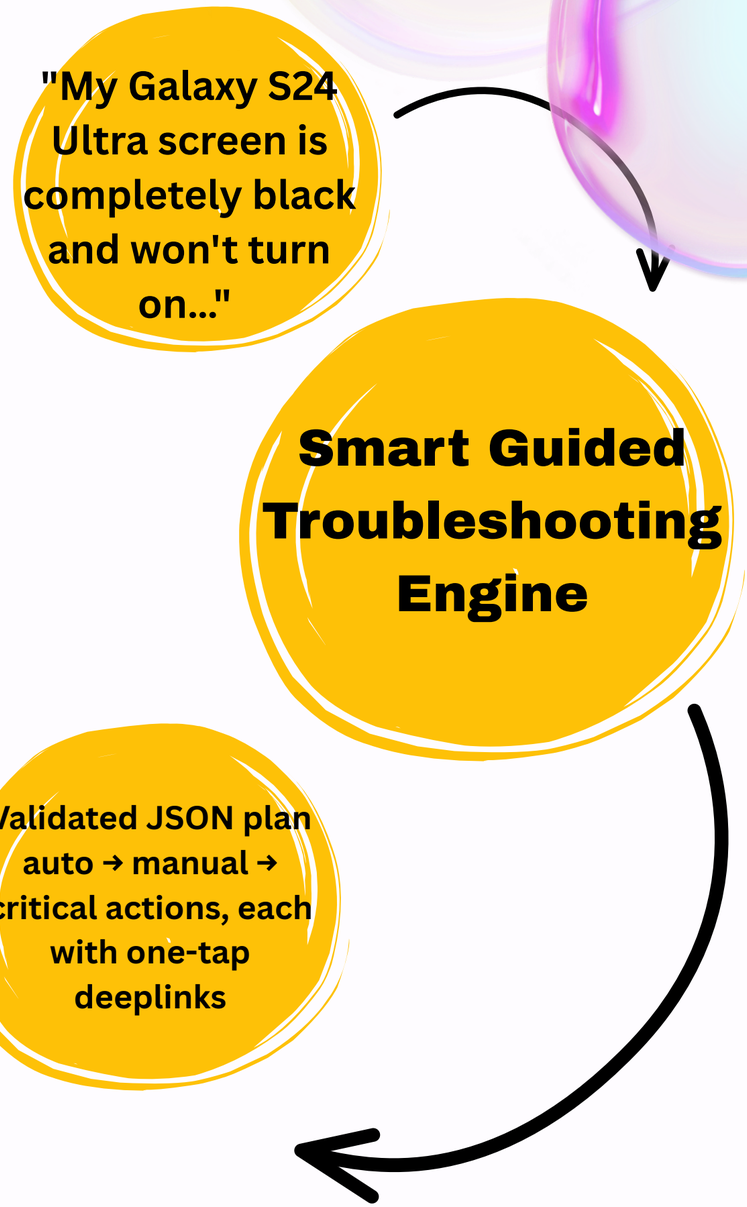
A vague complaint ("screen flickers and dies fast") is triaged by hand against a knowledge base, taking about 15 min per scenario. The customer then hunts through nested Settings menus.

**Goal**

Turn free-form text into an ordered, validated, one-tap plan: goal, title, categorised actions, steps and bixby:// deeplinks.

**Rule**

Never invent a fix. Steps come only from trusted Samsung (SIIS) text; deeplinks only from the 578-entry catalog; otherwise abstain.



# Existing solutions & gaps

| Present Today | Gap | Our Solution |
| --- | --- | --- |
| **Manual agent triage** | Slow (~15 min), inconsistent between agents, does not scale. | Instant, deterministic plan from the same trusted text. |
| **Keyword FAQ / support search** | Returns whole articles; customer still navigates Settings alone. | Step-level plan with one-tap Settings deeplinks. |
| **Plain LLM chatbot** | Can invent steps or links; no way to verify output. | LLM is a candidate only; catalog, URL-leak and schema checks gate it. |
| **Generic RAG** | Retrieves broad parent menus; ignores risk (reset may come first). | Parent-menu penalty and auto → manual → critical ordering. |

# Solution & architecture

**1. Complaint**
Free text (+ optional SIIS payload)

**2. Semantic cache**
Hit returns a validated plan, no LLM

**3. Query enrichment**
8–10 paraphrases; multiintent split

**4. Hybrid retrieval**
BM25 + BGE dense over chunked SIIS

**5. Tiered routing**
Exact / approx / web fallback / abstain

## Our Solution

**6. Extraction**
Gemini LLM, else deterministic parser

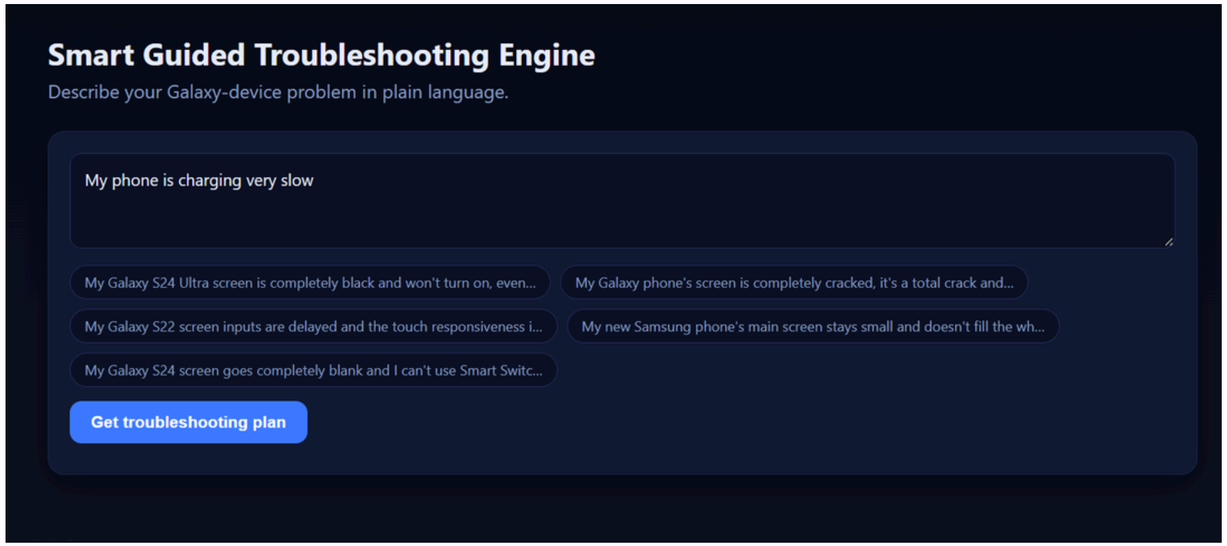
**7. Deeplink resolver**
5-term score; URI copied verbatim

**8. Risk sequencer**
Auto → manual → critical

**9. Validator + repair**
Pydantic, URL-leak, catalog, limits

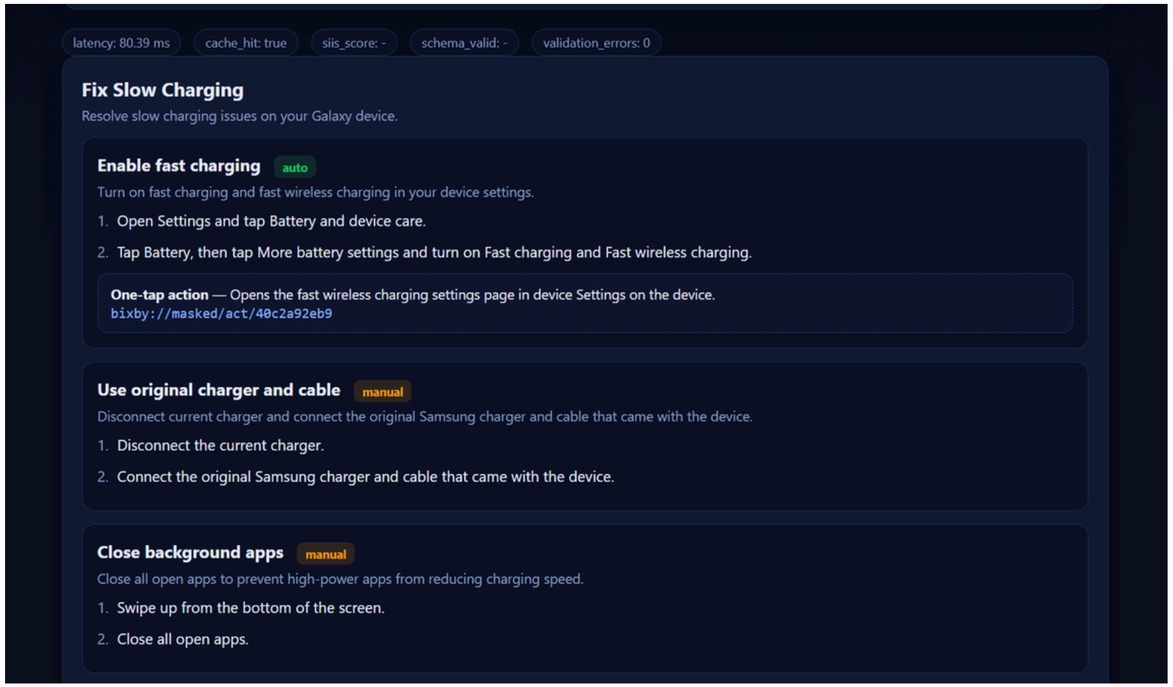
**10. Cache + JSON**
Plan, confidence, action graph

# Demo & product walkthrough




## INPUT

## OUTPUT



# Tools & tech stack used

## Backend & API

• Python 3.11, FastAPI, Uvicorn • Pydantic v2 response contract
• POST /v1/troubleshoot, GET /health

## Retrieval & ML

• rank-bm25 + BAAI/bge-small-env1.5
• FAISS / NumPy vector store
• Optional MiniLM cross-encoder reranker

## LLM (optional)

• Gemini via OpenAI-compatible API
• Structure extraction only
• Deterministic parser as fallback

## Web fallback & data

• Tavily, limited to samsung.com domains
• 578 masked Settings deeplinks
• Trusted SIIS support text

## Cache & infra

• Semantic cache: Jaccard + embeddings
• Redis 7 or in-memory
• Docker, docker-compose

## Frontend & QA

• Vanilla JS demo UI
• pytest, GitHub Actions CI, JSON logs

# Impact & use case

**~15 min**

manual triage per scenario (brief)

**1-3 s**

engine response after warm-up (run guide)

**22 ms**

median, offline path (measured)

### Agent assist

Samsung Care agents get a ready, ordered plan instead of reading articles.

### Self-service

Customers fix issues in-app with onetap deeplinks, no menu hunting.

### Safe automation

Reset and wipe steps are always last, so guided fixes stay non-destructive

# Innovation, results & limitations

## Innovation

- Verification-first: LLM output is only a candidate
- Verbatim deeplinks: LLM never sees the catalog
- Risk-aware order: reset always last
- Parent-menu penalty finds the exact screen
- Abstains instead of guessing

## Results

- 25/34 canonical queries resolve locally with verified deeplinks; 9 correctly abstain
- Local match 22 ms (p95 67 ms), Cold path p95 < 8 s, Cache hit < 5 ms
- All deeplinks verified against catalog

## Limitations

- Small corpus: Uncovered queries fall to Tavily
- Masked deeplinks: not testable on a real device
- Free LLM tier: Quota and latency vary
- Cold start: first query 30–60 s (model load)
- Free LLM tier limits latency and quota

# What's next

- Closed loop: run validationDeeplinks on-device, then advance or skip steps
- Live SIIS ingestion instead of a static knowledge set
- Extend beyond display, battery, connectivity, sound and camera
- Device and One UI version-aware plans

# Brownie points: why we stand out

✗ Can hallucinate steps and URLs

✗ Guesses an answer anyway

✗ One blended answer

✗ LLM call on every request

✗ Needs a hosted API



✓ Blocked by catalog and URL-leak checks

✓ Abstains and exposes a confidence score

✓ Handles multi-symptom complaints

✓ LLM fires only when parsing fails — cuts latency 80%

✓ Runs offline by default, degrades gracefully



# Thank You